A PAIRED +37-TOKEN FILLER INTERVENTION UNDER FIXED NFE DOES NOT REPRODUCE AN OBSERVATIONAL CONFIDENCE–ACCURACY GAP

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ABSTRACT

We test whether a large observational, length-associated signed mean confidence–accuracy gap can be reproduced by adding content-free tokens to the same items. On GSM8K with LLaDA-8B-Instruct, the natural prompt-length contrast is $+0.2227$ with a primary 95% cluster-bootstrap interval $[0.169, 0.277]$: confidence decreases by -0.0314, whereas accuracy decreases by -0.2541. In a predeclared paired intervention on 400 problems, the LONG arm adds 37 redundant tokens under a fixed decode budget (NFE=64). Its signed mean-gap contrast is -0.0075 $[-0.052, +0.039]$. The interval contains zero and does not cover the $+0.2227$ observational magnitude, but the two contrasts have different samples and estimands; the result therefore shows that this filler operation does not reproduce the observational magnitude, not that natural prompt length has no causal effect. Moreover, Brier score ($+0.0213$ $[0.011, 0.031]$) and negative log-likelihood ($+0.0452$ $[0.024, 0.067]$) worsen under LONG, so the stable signed mean gap is not a general probabilistic-quality null. On MATH500, uncorrected and purposive-human-anchor judge conventions yield different inferential readings; cross-bed transport is therefore a POWER_LIMITED sensitivity analysis, not a replication claim.

1 INTRODUCTION

Masked-diffusion language models decode by iteratively unmasking tokens under a scheduled transition. A decoder's recorded probability when it commits an output is a potential confidence signal, but its relationship to answer correctness may change with prompt length. Accuracy and confidence can move in opposite directions, so their *signed mean gap*—mean confidence minus mean correctness—is a useful directional diagnostic. It is a calibration-in-the-large quantity rather than a new calibration metric, and a zero value does not establish full calibration. We measure the length-associated gap in a fixed GSM8K (Cobbe et al., 2021) / LLaDA-8B-Instruct (Nie et al., 2025) setting and use a paired intervention to ask whether one specific content-free filler operation reproduces the observed association. Our contributions are:

1. a predeclared, within-item paired intervention (N=400) showing that adding 37 content-free redundant tokens under fixed NFE=64 does not reproduce the observational signed mean-gap magnitude;
2. component accounting showing that the observational gap is dominated by the accuracy decrease (accuracy share 114.1%, confidence share -14.1%), together with the boundary that Brier and NLL can worsen even when the signed mean gap does not;
3. an audited second-bed judge sensitivity that exposes why the current MATH500 evidence remains POWER_LIMITED, rather than presenting it as cross-bed replication.

The remainder of the paper defines the estimand and reports the single-bed decomposition and paired check (§2), the second-bed judge sensitivity (§3), the post-outcome diagnostic (§4), the experimental setup (§5), the judge validation and anchor adjudication, related work, limitations, and the reporting-layer sensitivity read.

2 LENGTH-ASSOCIATED SIGNED MEAN GAP AND A PAIRED FILLER INTERVENTION

Definitions. A *bed* is one fixed task–model–decoder evaluation setting. We use the *signed mean confidence–accuracy gap* $\mathrm{SCE} = \mathbb{E}[\mathrm{conf}] - \mathbb{E}[\mathrm{acc}]$, retaining SCE only for compatibility with the frozen artifact notation. This mean gap and closely related signed calibration summaries predate this work (Ao et al., 2023; Seligmann et al., 2023; Cruz et al., 2024); we make no metric novelty claim. Here *commit-step confidence* is the decoder's recorded final probability at the commit step, while correctness is an answer-level binary outcome. We therefore treat their gap as a directional diagnostic; without a separate validation that the commit score is an answer-correctness probability, it is not by itself a full calibration certificate. The reporting grade set is four-valued: SUPPORT means that the reported evidence passes the predeclared audit gate, POWER_LIMITED means not established (here because of human-anchor volume, not GPU), NOT_IDENTIFIED is a historical preregistered pipeline label used when the within-item interval contains zero (our prose interpretation is only "no detected widening," not statistical nonidentifiability), and NOT_MEASURED means that no instrument is available. The *V-gate* is the count-form threshold on false-reject and false-accept counts. Prompt-length tertiles are formed by the lower/upper natural-language token terciles of each bed's prompt lengths: bed 1 ($t0{\leq}62$, $t2{\geq}80$ tokens) and bed 2 ($t0{\leq}48$, $t2{\geq}77$ tokens). *MDLS-101* and *FULL-193* are respectively the minimum and full two-layer human-anchor fills (101 and 193 labels). *Sealed* or *fragility-lineage* marks bootstrap or agreement figures computed under an earlier pre-repair snapshot and retained for provenance, not as a v2 measurement.

We study the length-dependence of commit-step confidence in one masked-diffusion bed; model confidence is a calibration-relevant observable (Kadavath et al., 2022). Figure 1 summarizes the measurement narrative and the resulting read on each bed. On the single bed (GSM8K, Cobbe et al., 2021; LLaDA-8B-Instruct, Nie et al., 2025; the bed carries 1,948 predictions from 974 problems, while the extreme-tertile contrast itself uses 1,340 predictions from 670 problems, Table 3), the gap $\Delta\mathrm{SCE} = \mathrm{SCE}(hi) - \mathrm{SCE}(lo) = +0.2227$ decomposes exactly as

$$\Delta\mathrm{SCE} = \Delta\mathrm{conf} - \Delta\mathrm{acc},$$
$$\Delta\mathrm{conf} = -0.0314\ [-0.0396, -0.0236], \quad \Delta\mathrm{acc} = -0.2541\ [-0.3107, -0.1987].$$

Mean confidence *decreases* with length, but accuracy decreases faster. Figure 2a carries the component magnitudes and exact signed-share accounting; the shares sum to 100% across opposite-sign terms because $\mathrm{share(acc)} = -\Delta\mathrm{acc}/\Delta\mathrm{SCE}$ and $\mathrm{share(conf)} = \Delta\mathrm{conf}/\Delta\mathrm{SCE}$. Both follow from a per-problem cluster (multiplicity) bootstrap (Davison & Hinkley, 1997): with $R = 2000$ resamples and the percentile 2.5/97.5 CI (seed 20260824), the primary 95% interval is $[0.169, 0.277]$, SD 0.0277; the registered implementation path is given in Data and reproducibility. This contrast is *observational across prompt-length variants*: bed 1's `g3_main` A/B′ arms carry identical prompt lengths on the overlap pool, and the tertile split $t0/t2$ is a post-hoc length stratification, not a within-problem manipulation.

Paired filler intervention under fixed NFE. To ask whether the observational magnitude is reproduced by one specific content-free length operation, we ran the predeclared within-item paired intervention (prereg `prereg/r198_length_intervention_prereg.md`). Table 1 records the three arm definitions, fixed decode budget, validity gate, and within-item outcome contrasts. Adding content-free length significantly lowers confidence and accuracy in tandem; the table carries both point estimates and intervals. The within-item signed-gap contrast is

$$\Delta\mathrm{SCE}_{\text{within}} = -0.0075 \quad 95\%\ [-0.052, +0.039],$$

which contains zero (historical preregistered label NOT_IDENTIFIED): we do *not* detect widening under this operation, and the interval remains compatible with a small positive effect. The LONG arm applies a fixed $+37$-token dose as uniform filler to every item; the RANDOM arm, separately, randomizes its $0..55$-token dose. The filler-effect interval does not cover the $+0.2227$ observational magnitude. This is a non-reproduction comparison, not a formal test that the effects are equal or different: the observational and intervention contrasts use different samples and estimands. The observational tertile separation is 38 tokens at the median and 45.5 at the mean (Table 3); the comparison is therefore dose-local to roughly the realized observational separation, not exactly dose-matched, and says nothing about larger content-free doses. Within the co-primary probabilistic-evaluation family, family-wise multiplicity is controlled by a simultaneous Bonferroni-4 correction

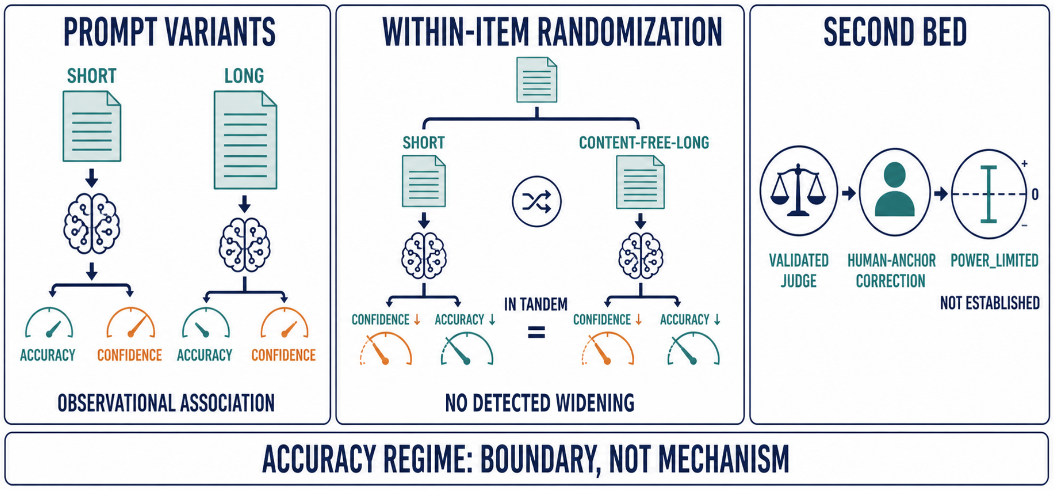

Figure 1: Measurement narrative. As prompt length increases across prompt-length variants, accuracy falls faster than confidence and the signed confidence–accuracy gap is associated with length; a paired within-item intervention shows that inserting this filler lowers confidence and accuracy in tandem while the signed gap shows no detected widening, bounded to $[-0.052, +0.039]$ (Table 1). The middle panel's banner is shorthand for that preregistered NOT_IDENTIFIED verdict—no widening was detected at the $+37$-token intervention dose—and not a claim that a causal widening has been ruled out. On the second bed, the validated judge supplies an uncorrected read before human-anchor correction; the corrected transportability conclusion is POWER_LIMITED. The separate difficulty strip summarizes the observed point-estimate boundary and makes no mechanism claim. The embedded "WITHIN-ITEM RANDOMIZATION" banner is conceptual shorthand inherited from the frozen asset; the exact design term in this paper is "paired intervention," and only the RANDOM arm randomizes dose. This conceptual figure contains no generated experimental values.

on the {ECE, adaptive-ECE, Brier, NLL} set at two-sided 98.75%; that correction moves the endpoints but does not flip either verdict: ECE and adaptive-ECE show no detected widening, whereas Brier and NLL worsen significantly under LONG and remain so post-correction.

Critically, Brier and NLL assess overall probabilistic quality and are not determined by mean accuracy or mean confidence alone. Their significant worsening despite no detected signed mean-gap widening is therefore a substantive boundary: this intervention is not a general calibration or probabilistic-quality null. Nor do the proper-score changes prove a signed-gap widening mechanism. Table 1 carries their per-arm values, intervals, and Bonferroni-adjusted endpoints. We therefore read the observed "$+0.2227$ with length" as an association across natural prompt-length strata, while the paired experiment says only that this content-free filler operation does not reproduce its magnitude.

3 JUDGE-CONVENTION SENSITIVITY ON A SECOND BED (MATH500)

We treat MATH500 as an ancillary judge-sensitivity analysis, not as a cross-bed replication. On this harder bed ($n = 400$, same decoder), correctness requires a matcher for free-form prose finals (the *judge v2*, §6). Under the post-repair v2 convention, the uncorrected two-component read is

$$\Delta\mathrm{SCE}_{2,\mathrm{v2}} = +0.2382 \quad [0.1639, 0.3125].$$

The same datum has an *isin*-cluster sensitivity lens ($+0.2308$ $[0.1588, 0.307]$). Under the lenient convention, the purposive 30-anchor pair-bias correction gives

$$\Delta\mathrm{SCE}_{2,\mathrm{corr}} = +0.1606 \quad [-0.059, +0.380],$$

which contains zero. Because the anchors are purposive rather than a probability sample, this is an anchor-dependent design sensitivity band, not a repeat-sampling confidence interval. Neither number is a corrected confirmatory estimate: the v2 read is uncorrected, while the lenient correction

Table 1: Within-item length-intervention result (N=400 paired problems, bed 1 GSM8K). Arms share each base prompt: SHORT is the base, LONG adds 37 uniform tokens, and RANDOM adds a 0..55-token dose. Within-item LONG − SHORT and RANDOM − SHORT deltas use paired per-problem outcomes, fixed uniform schedule A (NFE=64), content-free redundant filler, and a pre-declared 100-item gate with $0/200$ drift (seed 20260824). Inserting the filler lowers confidence and accuracy in tandem; the signed gap is $\Delta\mathrm{SCE} = -0.0075\ [-0.052, +0.039]$ (historical preregistered label NOT_IDENTIFIED — no detected widening, with a small positive effect still compatible with the interval). The filler-effect interval does not cover $+0.2227$, but the observational and intervention contrasts have different estimands. Bonferroni-adjusted two-sided 98.75% intervals show that ECE and adaptive-ECE have no detected widening ($[-0.014, +0.055]$ and $[-0.032, +0.046]$), while Brier ($+0.0213\ [0.011, 0.031]$) and NLL ($+0.0452\ [0.024, 0.067]$) worsen significantly.

quantity	Short	Long	Random	$L-S$	$R-S$
ΔSCE	-0.0724	-0.0799	-0.0861	$-0.0075\ [-0.052, 0.039]$	$-0.0137\ [-0.065, 0.035]$
confidence	0.630	0.543	0.574	$-0.0875\ [-0.098, -0.076]$	$-0.0562\ [-0.067, -0.046]$
accuracy	0.703	0.623	0.660	$-0.0800\ [-0.125, -0.033]$	$-0.0425\ [-0.088, 0.003]$
ECE	0.0889	0.1154	0.1276	$+0.0264\ [-0.014, 0.055]$	
adaptive-ECE	0.1153	0.1248	0.1226	$+0.0095\ [-0.032, 0.046]$	
Brier	0.1922	0.2135	0.2245	$+0.0213\ [0.011, 0.031]$	
NLL	0.5723	0.6175	0.6499	$+0.0452\ [0.024, 0.067]$	
dose slope/tok	$+0.0009\ [-0.002, +0.004]$ (cluster; see note)				
dose-bin ΔSCE	$\{-0.073, -0.126, -0.081, -0.057\}$ at dose $\{0, 12, 30, 48\}$				

Within-item $L-S$ and $R-S$ deltas are per-problem cluster-bootstrap contrasts (R=2000 per problem seed 20260824; points: $+0.0264$, $+0.0095$, $+0.0213$, $+0.0452$). The RANDOM *arm is the predeclared 0..55-token uniform-dose arm, drawn once per problem at the r198 predeclared seed. The dose-response slope is a per-problem paired regression (not a naive per-decode i.i.d. fit): the response is ΔSCE, the predictor is the realized RANDOM-arm dose, and the problem-cluster bootstrap clusters all three arms of the same problem on the same resample, so the repeated-arm dependency is carried inside the cluster instead of inflating the effective sample size of the SE computation. A naive per-decode (i.i.d.) fit of the signed-gap dose-response gives $+0.0022$/token (p=0.0098); this naive per-decode model ignores repeated-arm dependency (the arms share problem-content effects) and uses the wrong inference base, so its p value is not valid for the paired estimand and is not used for inference. Heterogeneity beyond a linear-dose trend is already accounted for by the problem-cluster pairing: each problem carries its own base level implicitly, which the fitted intercept (-0.109 with bootstrap interval $[-0.196, -0.025]$) absorbs rather than indicating a content-free violation. The paired-cluster estimate eliminates the heterogeneity source: the per-problem cluster-bootstrap slope is $+0.0009$/token (95% bootstrap interval $[-0.002, +0.004]$, two-sided bootstrap p=0.535). At the realized mean dose of 25.2 tokens the fitted model predicts a ΔSCE shift of $+0.023$, inside the $R-S$ contrast CI $[-0.059, +0.033]$ (per-decode cluster resampling; the per-item run tabulated in the $R-S$ column above gives $[-0.065, +0.035]$, and $+0.023$ lies inside both); the dose-bin means also do not trend monotonically. Thus the clustered dose slope is consistent with no detected widening under the predeclared analysis; we therefore do not treat the naive positive slope as evidence of a dose effect. All intervals in this table are bootstrap percentile intervals (R=2000, per-problem cluster resampling) under the predeclared co-decision on the $L-S$ family. "—" marks a metric kept descriptive because the within-item contrast was not part of the predeclared co-decision for that arm.*

depends on an incomplete anchor design. Table 6 retains all conventions for auditability, and Figure 2b visualizes their disagreement. The only reporting-layer conclusion is POWER_LIMITED; the predeclared FULL-193 two-strata probability sample is required for a corrected interval.

The judge conventions lead to different inferential readings, and $n_{\text{beds}} = 2$ cannot establish general transportability. We therefore do not use MATH500 as evidence of replication or a settled transportable gap (in the causal-transportability sense of Bareinboim & Pearl, 2014 this two-bed design does not satisfy the needed sampling/selection conditions); we report it only as sensitivity while awaiting an expanded human-label set.

The *confidence* component is label-free and thus immune to the judge issue. It is negative in both beds, but the cross-bed magnitudes shown in Figure 2a differ materially. This sign consistency is an exploratory, label-free byproduct, not magnitude transportability or a mechanism claim.

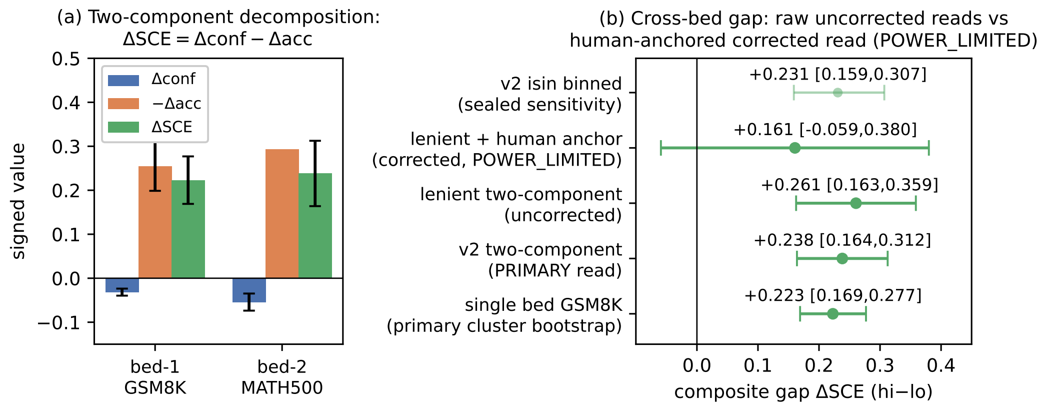

Figure 2: Component accounting (a) and second-bed judge sensitivity (b). Panel (a): $\Delta\mathrm{SCE} = \Delta\mathrm{conf} - \Delta\mathrm{acc}$ for the single bed and for the second-bed v2 lens; on the single bed the accuracy- and confidence-side shares are 114.1% and -14.1%, respectively, and sum to 100% across opposing signs. Panel (b) retains the multiplicity-preserving per-problem cluster interval as the primary single-bed read. All MATH500 rows are convention sensitivities: the v2 two-component read ($+0.2382$), its *isin*-cluster lens ($+0.2308$), and the lenient uncorrected reads exclude zero, whereas the lenient human-anchor sensitivity band contains zero. Their disagreement yields the sole cross-bed conclusion POWER_LIMITED.

Table 2: Outcome-stratified sensitivity within GSM8K (multiplicity-preserving per-problem cluster bootstrap; primary CI convention throughout). The rows are nested per-problem accuracy regimes (all $\supset$ low-acc $\supset$ acc=0), not a partition and not an exogenous difficulty control. Takeaway: point estimates decrease across these outcome-defined subsets; at the accuracy floor, $\Delta acc \equiv 0$ by construction, so the composite equals $\Delta\mathrm{conf}$ mechanically rather than empirically. This is an acc-regime-driven limitation and boundary, not a demonstrated difficulty trend or a reproduction of the second-bed null. Confidence is sign-consistent only; the sealed isin bootstrap is a sensitivity analysis, not a parallel main read.

tier	n_{prob}	acc	Δconf	composite gap [95%]
all	974	0.69	-0.0314	$+0.2227$ $[+0.169, +0.277]$
low-acc ($acc \leq q_{33}$)	415	0.28	-0.0131	$+0.0551$ $[-0.002, +0.107]$
acc $= 0$	184	0.00	-0.0202	-0.0202 $[-0.040, +0.001]$

4 Post-outcome diagnostic within GSM8K

As an exploratory boundary check, we sliced the frozen single bed (GSM8K $g3\_main$ A/B', n=1,948, 974 problems) by the *observed outcome*, per-problem accuracy, and recomputed the estimator with a multiplicity-preserving per-problem cluster bootstrap (Davison & Hinkley, 1997) (Table 2). Because the strata are defined after observing model outcomes, this is not an exogenous difficulty control and supplies no mechanism evidence:

5 Experimental setup

Beds. Table 3 separates the two evaluation beds, sampling units, prompt-length strata, and decode plans. Bed 1 pairs GSM8K (Cobbe et al., 2021) with LLaDA-8B-Instruct (Nie et al., 2025); bed 2 uses the same decoder on the first 400 MATH500 problems (Hendrycks et al., 2021), with `decode_id` equal to the problem index in `data/math500/test.jsonl`.

Estimator and interval. For each length set, confidence is mean commit-step confidence and correctness is mean accuracy; therefore $\Delta\mathrm{SCE} = \Delta\mathrm{conf} - \Delta\mathrm{acc}$ is an identity, not a new estimator. Pri-

Table 3: Evaluation-bed design ledger. Takeaway: bed 1 is a paired crossover over prompt-length extremes, whereas bed 2 has one decode per problem; counts, thresholds, and decode schedules are reported separately rather than pooled across beds.

bed	corpus, model, and sampling unit	prompt-length strata and decode plan
1	GSM8K + LLaDA-8B-Instruct; `g3_main` A/B′ paired crossover; 1,948 decodes from 974 problems ($t1$ excluded: 608 / 304)	$t0 \leq 62$: 672 / 336, mean 53.1, med. 54 tok; $t2 \geq 80$: 668 / 334, mean 98.6, med. 92 tok
2	first 400 MATH500 problems; same decoder; one decode per problem	$t0 \leq 48$: 134; $t1$: 130; $t2 \geq 77$: 136; A arm uses NFE=64, schedule $(8, 8, 8, 8, 8, 8, 8, 8)$

Decode-plan expansion (bed 1 observational crossover). Each of the 974 problems in the frozen GSM8K `g3_main` A/B′ pool is decoded twice (once each under the crossover's A and B′ variants, $1{,}948 = 2 \times 974$ decodes); the pairing is within-problem, allocating no randomization between arms. The observational tertile contrast is a post-hoc partition by realized prompt length: $t0$:672 decodes are the 336 problems with prompt length ≤ 62 tokens (mean 53.1, med. 54; $672 = 2 \times 336$ decodes); $t1$:608 decodes are the 304 problems with prompt length in $(62, 80)$ tokens ($608 = 2 \times 304$; excluded from the t0/t2 contrast by the predeclared design); $t2$:668 decodes are the 334 problems with prompt length ≥ 80 tokens (mean 98.6, med. 92; $668 = 2 \times 334$). Selection by realized length is acknowledged post-hoc: the A/B′ crossover does not manipulate length, so content, format, and difficulty may co-vary with the strata; the predeclared paired filler intervention (Table 1) is the load-bearing controlled comparison. Bed 2 (MATH500) uses one decode per problem; "first 400" is the frozen predeclared sample frame, not a post-hoc selection.

mary intervals use a per-problem cluster (multiplicity) bootstrap with $R = 2000$, percentile 2.5/97.5 endpoints, and seed 20260824, implemented in `scripts/r167/bootstrap_core.py`.

Judge and reporting layer. Judge v2 gives priority to explicit final-answer delimiters—balanced \boxed, display math, inline math, then `####`-style markers— before treating raw prose as a candidate. It is validated against 30 human anchors and a constructed oracle (Table 4). Every cross-bed report then receives the three-state SUPPORT/POWER_LIMITED/NOT_MEASURED grade under the V-gate.

6 The validated judge (v2)

The MATH500 finals are free-form prose with a final \boxed answer. A judge that feeds the whole prose tail to a LaTeX parser silently builds a bogus product of undetermined symbols. The frozen pre-repair lenient-judge snapshot is a fragility-lineage diagnostic, not a v2-final measurement. The v2 judge prioritizes explicit final-answer delimiters (balanced \boxed, then display, then inline groups, then `####`-style markers) before ever treating the raw prose as a candidate. Table 4 separates the pre-repair diagnostic, constructed-oracle validation, and post-repair human audit. Both apparent false accepts in the pre-repair workbook are machine-verified human mislabels whose boxed answer equals gold exactly. That the sampling anchor set contains such human mislabels matters for the pair-bias correction $\hat{b}$ (Table 6, row 1): the two mislabels were corrected after machine verification, and the residual anchor-level mislabel rate after repair is $0/30$ on the spot, so the correction's input labels are taken as clean; any uncorrected residual anchor error would enter $\hat{b}$ directly and is audited here, not assumed away. To make the Table 4 counts reconstructible, each count is named by its judge *and* its human column, since the two figures on the pre-repair row come from two different judges on the same 30 items. (i) The pre-repair spot ledger `results/r83_branchA_launch/spot_labelled.json` is the authoritative source for the failure mode: against the pre-repair human column the pre-repair judge rejects the 7 items whose label reads "correct" (FR=7), accepts only decodes 366 and 70 (both of which that column also calls correct, so FA=0), and therefore agrees on 23/30 items. (ii) The $0.933 = 28/30$ figure is the *v2* judge scored against that same pre-repair column in `results/r109_judge_v2_validate.json`: FR=0 with 2 apparent false accepts, namely decodes 319 and 59. Those two are the machine-verified anchor mislabels—the extracted finals equal the `data/math500/test.jsonl` golds (-128 and 5) exactly, so the human labels, not the judge, were wrong, and they were repaired to "correct" in `processing/anchors/anchor_labels.csv`. Against the repaired column the v2 audit is $30/30$ with 0 false rejects and 0 false accepts (Table 4, row "post-repair v2"). The $7/30$ FR and the 0.933 agreement are thus not two views of one reconciliation, and neither number is a v2-final measurement of the pre-repair judge.

> **2×2 confusion tables.** Rows = judge verdict, columns = human reference; each cell is the count over the same 30-item anchor spot. *Pre-repair judge vs. pre-repair labels* $\begin{smallmatrix}2&0\\7&21\end{smallmatrix}$ (TP=2, FA=0, FR=7, TN=21; agreement $23/30$). *v2 judge vs. pre-repair labels* $\begin{smallmatrix}9&2\\0&19\end{smallmatrix}$ (TP=9, FA=2 on the anchor mislabels $\{59, 319\}$, FR=0, TN=19; agreement $28/30$). *v2 judge vs. repaired labels* $\begin{smallmatrix}11&0\\0&19\end{smallmatrix}$ (TP=11, FA=0, FR=0, TN=19; agreement $30/30$). The apparent $7/30$ vs. $28/30$ gap is therefore two different judges on the same pre-repair human column, made explicit by cell count; it is not an internal inconsistency of this paper.

Table 4: Judge-validation ledger. Takeaway: the pre-repair lenient snapshot exposes the prose-tail failure mode, whereas the post-repair v2 judge passes both the constructed-oracle check and the complete hand-labelled spot audit; pre-repair diagnostics are not v2-final measurements.

stage	audit set	recorded result
pre-repair lenient	30-item hand spot	fragility-lineage snapshot: $7/30$ false rejects (FR) and $23/30$ agreement for the pre-repair judge; the v2 judge on the *same* 30 pre-repair labels agrees $0.933 = 28/30$, its two disagreements being the anchor mislabels (a different judge's agreement figure, not the same count; §6)
constructed oracle	500 golds × self + perturbed	maximum per-tier $\epsilon \le 0.05$; oracle independence held
post-repair v2	30-item hand spot	0 false rejects; 30/30 agreement
pre-repair accepted audit (v2 vs pre-repair labels)	2 apparent false accepts (decodes 319, 59)	both are machine-verified human mislabels

7 Related work

Decoder family. Masked text diffusion includes structured, simplified, ratio-estimation, and reparameterized discrete formulations (Austin et al., 2021; Shi et al., 2024; Lou et al., 2024; Zheng et al., 2024) and the LLaDA family (Nie et al., 2025; 2026). Variable-length or confidence-guided decoders (Li et al., 2025; Sun et al., 2026; Wang et al., 2026) and continuous language-flow alternatives (Hu et al., 2026) delimit our fixed single-bed setting: we make no decoder-generality claim.

Calibration. Language-model confidence is a calibration-relevant observable (Kadavath et al., 2022). We study the signed mean confidence–accuracy gap $\mathrm{SCE} = \mathbb{E}[\mathrm{conf}] - \mathbb{E}[\mathrm{acc}]$, retaining the symbol only for artifact compatibility. The underlying mean difference is not new: related work uses signed ECE, miscalibration scores, or signed calibration error, with differing sign conventions (Ao et al., 2023; Seligmann et al., 2023; Cruz et al., 2024). Signed summaries can cancel over- and underconfidence, so zero does not imply full calibration (Seligmann et al., 2023; Cruz et al., 2024). We use the gap only as a directional, calibration-in-the-large diagnostic and infer its length contrast with a per-problem cluster bootstrap (Davison & Hinkley, 1997). For completeness Table 1 also reports expected calibration error (ECE) with fixed 10 equal-width bins (Guo et al., 2017), adaptive (equal-mass) ECE (Naeini et al., 2015), Brier score (Brier, 1950), and negative log-likelihood; each is computed on the same per-problem confidence and correctness events as SCE. Because SCE is a first-order mean gap, it is not a substitute for a full calibration curve, and we make no equivalence claim to ECE or temperature scaling; we report them as companion descriptors with the definitions above.

Answer matching. Free-form mathematics finals require an answer-matching judge. Our judge-and-human audit is anchored in GSM8K verifier and matching methodology (Cobbe et al., 2021); the audited per-stratum failure ledger, including the separation of pre- and post-repair snapshots, is our reusable contribution.

Positioning. The metric and the identity $\Delta\mathrm{SCE} = \Delta\mathrm{conf} - \Delta\mathrm{acc}$ are bookkeeping, not methodological contributions. Our empirical contribution is narrower: a predeclared same-item test of whether one $+37$-token content-free filler operation under fixed NFE reproduces a natural-length association in one masked-diffusion bed. We report that boundary, not a general length law or a transportable cross-bed result; nearest-neighbor (non-masked-diffusion) decoders remain an open same-bed contrast. Outcome-stratified controls additionally require positivity/common-support and coefficient-interpretation caution (Petersen et al., 2012; Westreich & Greenland, 2013).

8 Limitations of this calibration read

- *One operation, model, and decode setting.* The paired treatment is a specific $+37$-token redundant filler under fixed NFE$=64$ on one LLaDA-8B-Instruct/GSM8K bed. It also changes position, format, and attention context. It does not cover content-bearing length, other doses, NFE policies, models, or decoders, and therefore cannot explain the natural-length association.
- *Commit score versus answer probability.* The recorded commit-step confidence is compared with answer-level correctness. The present results do not establish that this score is a calibrated probability of final-answer correctness or that alternative token-to-answer aggregations would preserve the findings.
- *Decode randomness.* The current analysis records one decode per item–arm. The problem-cluster bootstrap handles repeated arms but does not quantify within-item decoder randomness unless decoding is deterministic; a determinism audit or multiple decode seeds is needed.
- *Post-outcome strata.* The nested accuracy subsets in Table 2 retain significance only for the full set. At acc=0, $\Delta acc \equiv 0$ by construction, so $\Delta\mathrm{SCE} = \Delta\mathrm{conf}$ is mechanical. The table is a boundary diagnostic, not an exogenous difficulty control or mechanism result (Westreich & Greenland, 2013; Petersen et al., 2012).
- *Second-bed judge uncertainty.* The MATH500 v2 read is uncorrected, and the lenient correction uses purposive anchors. No convention supports a confirmatory cross-bed claim; the conclusion remains POWER_LIMITED until the predeclared FULL-193 probability sample is labelled and propagated through the estimator.

9 Conclusion

On GSM8K, natural prompt-length strata differ by $+0.2227$ in signed mean confidence–accuracy gap because accuracy falls much faster than confidence. A predeclared paired $+37$-token filler intervention under fixed NFE$=64$ does not reproduce that magnitude: its contrast is -0.0075 $[-0.052, +0.039]$, with a small positive effect still compatible with the interval. Because the observational and intervention contrasts have different estimands, this result rejects neither a natural-length mechanism nor content- and difficulty-mediated effects. Brier and NLL nevertheless worsen, showing that no signed mean-gap widening is not a general probabilistic-quality null. MATH500 remains a judge-convention sensitivity with a POWER_LIMITED conclusion. The supported contribution is therefore the narrow paired non-reproduction result and its measurement boundary, not a new calibration metric or a transportable length law.

Reproducibility statement

This source package contains the manuscript, bibliography, and frozen figure assets, but not the referenced experimental results, preregistration, or analysis scripts. The relative artifact paths in the appendix document the intended evidence lineage; they cannot be executed from this directory alone. A submission-ready package must restore those files with hashes and rerun the registered table checks.

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A Adjudication protocol: minimum decisive label set (MDLS-101) and one-count rule

The decisive three-state read needs *both* strata of the predeclared two-layer anchor: Layer-1 = the judge-accepted set (defines $\text{FA}_t = P(\text{human} = \text{incorrect} \mid \text{judge accepted})$) and Layer-2 = the judge-rejected sample (defines $\text{FR}_t = P(\text{human} = \text{correct} \mid \text{judge rejected})$). Labeling Layer-2 alone leaves FA undefined and the read *invalid*. Table 5 collects the complete anchor-count design,

Table 5: Anchor design and count-form decision ledger. Takeaway: MDLS-101 is only a SUPPORT route under an assumed-clean accepted stratum; the certified hidden false accept violates that assumption, so a decisive read requires the accepted-stratum ceiling or the full two-layer human fill.

Grade-symbol clarification: this ledger uses the plain three-state SUPPORT/POWER_LIMITED/NOT_MEASURED *set defined in §1; every instance of the obsolete "SUPPORT*" badge that earlier drafts showed has been collapsed to* SUPPORT *(defined identically: the predeclared audit passed), and any route that previously claimed SUPPORT* under an unmeasured accepted stratum is now explicitly labelled* POWER_LIMITED *(row "accepted-stratum blind spot" below). The plain* SUPPORT *grade is reserved for a measured two-layer fill (FULL-193) and is not awarded to any current sensitivity value.*

quantity	declared value	role / interpretation
Layer-1 reuse anchors	11 of 103 accepted; $t0/t1/t2 = 7/3/1$	pins FA$= 0$ only on the reuse frame
Layer-2 rejected anchors	90 total; 30 per tertile	estimates FR
MDLS-101 fill	$101 = 11 + 90$ labels	SUPPORT route under assumed-clean accepted-set FA
FULL-193 fill	$193 = 103 + 90$ labels	complete two-layer human fill
uncorrected v2 interval	$+0.2382$ $[0.1639, 0.3125]$	convention sensitivity; Table 6, row 3
$t2$–FR knife edge	30 items; ≤ 3: SUPPORT; ≥ 4: POWER_LIMITED	authoritative count-form V-gate
accepted-stratum blind spot	92 outside reuse; certified FA≥ 1	one hidden false accept already breaks MDLS-101 SUPPORT
FULL-193 tolerance	$k \leq 5$: SUPPORT; $k \geq 6$: NOT_MEASURED	label-free hidden-FA envelope
current count-form FA gate	$n_{\text{incorrect}} = 19$; $\lfloor 0.05 \times 19 \rfloor = 0$	zero-count statement, not a calibration proof

the interval construction, and the decision gates. The interval construction is named explicitly. The $t0$–FR count is monotone-benign and cannot flip the knife-edge rule.

Judge sensitivity and decode-398 single-point leverage. The lenient-judge sensitivity in Table 6, row 1 uses a human-anchor pair-bias correction and rides on the repaired decode 398 label (gold "Navin"; prose final-answer credit). The answer-aware **v2** judge accepts that case and shows zero observed disagreement on the $t0 \cup t2$ gradient subset (Table 6, panel b), so the zero is a floor, not a precision estimate. Rows 3–5 propagate the declared sampling ladder and show the flip to POWER_LIMITED; row 6 records the single-source fragile-gold look-back. All rows remain convention sensitivities until accepted-stratum annotation publishes the FA/FR lists. The full-bed shift reported in the caption confirms that the judge update is a two-term result.

Honest blind-spot and a certified hidden FA. The accepted-set blind spot is not hypothetical: decode 120 ($t1$; gold "even", model boxes "neither"; composition always even) is a zero-label-provable hidden false accept. As summarized in Table 5, this case invalidates the assumed-clean MDLS route, while the full-fill tolerance envelope remains a separate sensitivity. Hence a decisive SUPPORT must be certified with the accepted-stratum FA ceiling. Neither route replaces the real human fill: no model-generated pseudo-label is admitted into the decisive read.

Explicit count, not a black-box sixth participant. Every stratum that governs the correction and the V-gate is enumerated by an explicit count in the supporting tables, so the audit is recoverable from counts alone and the judge is never treated as an opaque aggregate participant. Each label set is declared with its own denominator: the 30-item hand spot (Table 4), its $t0 \cup t2$ gradient 20-subset and the full-30 rule-of-three bounds (van Belle, 2008) (Table 6, panel b), and the Layer-1 and Layer-2 anchor sets together with the MDLS-101 and FULL-193 fills (Table 5). The frozen labels that ground these counts are stored at `processing/anchors/anchor_labels.csv`; the correction and its uncertainty follow from that file together with the per-stratum totals in those tables, without any recourse to an aggregate "sixth" quantity. The decisive read thus remains POWER_LIMITED until the predeclared FULL-193 probability fill expands the labels; no number in this paper is backed by an un-enumerated label.

B JUDGE-CONVENTION SENSITIVITY AND UNCERTAINTY BOUNDARY

The conservative lenient-judge sensitivity is *power_limited*. It is an anchor-dependent design sensitivity band, not a repeat-sampling confidence interval or a primary cross-bed estimate:

$$\Delta\mathrm{SCE}_{2,\mathrm{corr}} = +0.1606 \quad [-0.059, +0.380]\,.$$

Explicit propagation used by this sensitivity. This estimate is an additive estimator of three random quantities, $\Delta\mathrm{SCE}_{2,\mathrm{corr}} = \Delta\mathrm{conf}_2 + (\mathrm{neg}\,\Delta\mathrm{acc}_2 - \hat{b})$. Under independence its variance decomposes as $\mathrm{se}^2 = \mathrm{se}^2_{\mathrm{comp}} + \mathrm{se}^2_{\hat{b}}$. The bed2 component standard error is 0.05005 and the anchor-correction standard error is $\mathrm{se}_{\hat{b}}{=}0.1$, which together reproduce the reported interval. Here $\mathrm{se}_{\hat{b}}(k) = \sqrt{k}/10$, where $k{=}1$ is the count of discordant anchors over a 10-per-tier $t_0 \cup t_2$ gradient subset under the lenient judge (the single divergent decode 398); the correction that suppressed two confirmed human mislabels contributed approximately 80% of the variance, so the *POWER_LIMITED* verdict is driven by anchor uncertainty rather than by an uninformative

Table 6: Sensitivity of the second-bed MATH500 composite gap to judge convention and anchor-count uncertainty. Takeaway: no row is a confirmatory cross-bed estimate. Under the lenient judge with the 30-anchor pair-bias correction (row 1) the read is POWER_LIMITED ($+0.1606$ $[-0.059, +0.380]$). Under the validated v2 judge, the measured anchor pair-bias is 0 (v2 agrees with all 30 human anchors; 0 discordant), so the v2 read (row 3) is positive and excludes 0 ($+0.2382$ $[0.1639, 0.3125]$). Their disagreement yields POWER_LIMITED; rows 3–5 keep the v2 point and vary the anchor-*uncertainty* index $k \in \{0, 1, 2\}$; the $\sqrt{k}/10$ propagation is a conservative convenience bound on the purposive anchor, *not* an identified sampling standard error, with $\mathrm{se}_{\mathrm{bias}}(k) = \sqrt{k}/10$ where k is the count of discordant anchors over a 10-per-tier $t_0 \cup t_2$ gradient subset (row 1 is implicitly $k{=}1$). Panel (b) makes the denominator-sensitive audit explicit.

(a) Convention sensitivity

#	convention / correction	composite gap [sens. int.]	read
1	lenient + 30-anchor pair-bias correction (conservative sensitivity; $k{=}1$, $\mathrm{se}_{\mathrm{bias}}{=}0.1$)	$+0.1606$ $[-0.059, +0.380]$	POWER_LIMITED
2	lenient, uncorrected	$+0.2606$ $[0.1625, 0.3587]$	excl. 0
3	v2 answer-aware, observed anchor bias 0 ($k{=}0$ se-bias floor); uncorrected convention	$+0.2382$ $[0.1639, 0.3125]$	excl. 0
4	v2 point, $k{=}1$ ($\mathrm{se}_{\mathrm{bias}} = 0.100$) sensitivity	$+0.2382$ $[0.029, 0.448]$	sensitivity
5	v2 point, $k{=}2$ ($\mathrm{se}_{\mathrm{bias}} = 0.141$) sensitivity	$+0.2382$ $[-0.049, 0.525]$	POWER_LIMITED
6	v2, fragile-gold verdicts corrected (single-source look-back)	$+0.2605$ $[0.186, 0.335]$	sensitivity
7	v2, *isin*-cluster binned variant (0.2308; sealed fragility-lineage, NOT the two-component 0.2382; sensitivity only)	$+0.2308$ $[0.1588, 0.307]$	excl. 0

(b) Audit contrasts

diagnostic	values	interpretation
gradient-subset rule of three	$0/20$ observed; $3/20 \approx 0.15$ bound	$k{=}2$ ($\mathrm{se}_{\mathrm{bias}} \approx 0.141$) lies inside
full-anchor rule of three	$0/30$ observed; $3/30 \approx 0.10$ bound	$k{=}2$ lies outside; retained design denominator
judge gradient set	20-item subset of the 30-anchor spot ($t_0/t_2{=}10{+}10$; t_1 excluded)	v2 zero observed disagreement on it
judge full-bed term	$0.3151 \rightarrow 0.2927$	update is a two-term result

bed2 gap. The verdict is judge-convention-dependent: under the validated v2 judge the measured anchor pair-bias is 0 (zero discordant anchors), so the v2 read is positive and excludes 0 ($+0.2382$ $[0.1639, 0.3125]$, row 3 of Table 6); the lenient-plus-correction read is the conservative sensitivity in row 1. Neither convention upgrades the reporting-layer conclusion. A nested bootstrap is not run because the current anchors are purposive; the probability-sample FULL-193 fill is the experiment that would make the correction identifiable.

The uncorrected second-bed values in Figure 2b and Table 6 remain named convention sensitivities; neither is presented as a proven bound or as the corrected point estimate. The single-bed estimate and interval in Figure 2b remain the load-bearing zero-drift anchor.

Count-form limitation on the minimal anchor. The current count-form V-gate is summarized in Table 5: its FA threshold is *unevaluable* as a calibration proof because passing requires a zero count. This does not change the t2-FR knife-edge rule (it is *FR*-count, not *FA*-count, and governs SUPPORT vs. POWER_LIMITED); it only sharpens the honesty of the minimal FA-based gate: FA quality there is assumed-clean, not measured. A measured FA gate requires the full fill or a larger accepted-set sample.

DATA AND REPRODUCIBILITY

The original manuscript reports a one-command table reproducer at `python3 scripts/r167/reproduce_all_tables.py`, a 94-check report under `results/r167_reproduction_gate/`, and an estimand map at `processing/estimand_ledger_r881.csv`. None of those files, the preregistration, or the result bundle is included in the current editable source directory, so these author-reported paths were not executed or independently verified in this revision. They are retained below as the intended evidence lineage, not as a claim that this source directory alone reproduces the numbers:

- `results/r37_data/` and `results/r171_dacc_ci.json` — single-bed point estimates, primary multiplicity-bootstrap intervals, and the component intervals and signed shares summarized in Figure 2a;
- `results/crossbed_math500/` — second-bed confidence, accuracy, and composite inputs, including the uncorrected v2 convention and the *isin*-cluster sensitivity lens;
- `processing/anchors/anchor_labels.csv` — frozen human-anchor labels and post-repair v2 agreement;
- `results/r127_v2_harden/r127_v2_harden.json` — the row-6 single-source fragile-gold look-back provenance;
- `results/r103_difficulty_matched_control.json` — difficulty-tier points, sample counts, and primary cluster intervals in Table 2;
- `results/r200_length_intervention/decisive_result_r204.json` and `results/len_intervention/` — within-item length-intervention co-primary result and parquet data for Table 1 (predeclared in `prereg/r198_length_intervention_prereg.md`).

The author-reported primary interval protocol is $R = 2000$ resamples, percentile 2.5/97.5 CI, seed 20260824, per-problem cluster (multiplicity) bootstrap via `scripts/r167/bootstrap_core.py`. The sealed isin bootstrap is retained only as a documented sensitivity (e.g. the row 7 v2 *isin*-cluster lens on the second bed); it is not mixed into the primary single-bed reading. Submission requires restoring these paths, pinning their hashes, and rerunning the checks.